

MAHAM FAIZ MALIK

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EDUCATION

BS Physics

Sept 2021 - July 2025

PIEAS - Pakistan Institute of Engineering and Applied Sciences, Islamabad

Courses related to Physics: Mathematical Methods of Physics, Quantum Mechanics, Classical Mechanics, Electromagnetic Theory, Electronic Circuit Design, Electronic Devices and Circuits, Thermal and Statistical Physics, Modern Physics, Optics, Heat and Thermodynamics, Basic Nuclear Engineering, Particle Physics, Quantum Field Theory

Courses related to Mathematics: Calculus, Linear Algebra, Mathematical Methods of Physics, Probability and Statistics, Ordinary Differential Equations

Courses related to Computer Science: Computer Fundamental and Programming, Intro to Machine Learning

CGPA: 3.50/4.00

TECHNICAL SKILLS

Programming: Python, C

Python Libraries: Lyapunov, Tensorflow, Cirq

Typesetting and Drawing: Latex, Microsoft Office

WORK EXPERIENCE

Intern (PTPRI - Pakistan Tokamak Plasma Research Institute)

July 2023 - August 2023

Project : Analytical Analysis of Transient Heat Transfer

- Modeled transient heat transfer in plasma gasification chamber systems and comparison with ANSYS.
- Learned plasma physics fundamentals and reactor material. behavior.

Intern (NILOP - National Institute of Laser and Optonics)

JULY 2024 - August 2024

Introductory Quantum lab experiments

- Performed quantum optics and laser experiments.
- Conducted single-photon and optical detection measurements.

PROJECTS

Stand-Alone PV Water-Pumping System (Theoretical System Design Project)

November 2025

- Completed the full design of a standalone photovoltaic water-pumping system, performing solar irradiance assessment, hydraulic load analysis, PV array sizing, component selection, and performance simulations for off-grid operation.

Entanglement in Coupled Cavity Optomechanics (Final Year Project)

Sept 2024 - July 2025

- Conducted an in-depth study of optomechanical systems, analyzing how energy variations affect the system Hamiltonian. Focused on entanglement in tripartite optomechanical systems via optical-dark-mode control, investigating how breaking the optical dark mode enhances entanglement. Reproduced and validated all results from the research paper *Tripartite Optomechanical Entanglement via Optical-Dark-Mode Control* (Deng-Gao Lai et al., RIKEN & Univ. of Michigan, 2023), theoretical models and simulations using Python Lyapunov libraries.

Quantum-Assisted Binary Image Classification (Machine Learning Project)

2025

- Implemented and evaluated a quantum-assisted binary image classification model using TensorFlow Quantum and Cirq, improving accuracy through hybrid quantum-classical circuits and gaining hands-on experience with quantum machine learning and parameterized quantum circuit design.

Data Analysis Project (Semester Project) 2023

- Survey Analysis of Anxiety Levels in University Students. Performed statistical analysis with Python.

Structure of Ceramics (Semester Mini-Research Project) 2022

- Literature study on crystallography and structure–property relations.

Seat Allocation System for Pakistan Eagle Flight PE-49 Using C Language (Semester Project) 2022

- Development of software using C language to implement a seat allocation system for Pakistan Eagle Flight PE-49, handling seat selection, fare calculation, and displaying messages based on seat availability and class type.

ATM Machine Implementation Using C Language (Semester Project) 2022

- Development of a C language code for the implementation of ATM machine.

WORKSHOP

Quantum Workshop (PIEAS) February 17 - 21, 2025

- Participation in international workshop on Recent Advances in Quantum Science and Technology.

CERTIFICATES AND ACHIEVEMENTS

Best Final Year Project Certificate in Bachelors (ORIC - PIEAS) 2025

Certificate of Appreciation for winning Cricket in Sports Week (PIEAS) 2023

Certificate of Appreciation for winning Tug f War in Sports Week (PIEAS) 2023

LANGUAGES

English, Fluent